Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Lab Assessment 5

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Course title: Agile Development Process and Devops Lab

Slot: L57+L58

Course code: ISWE406P

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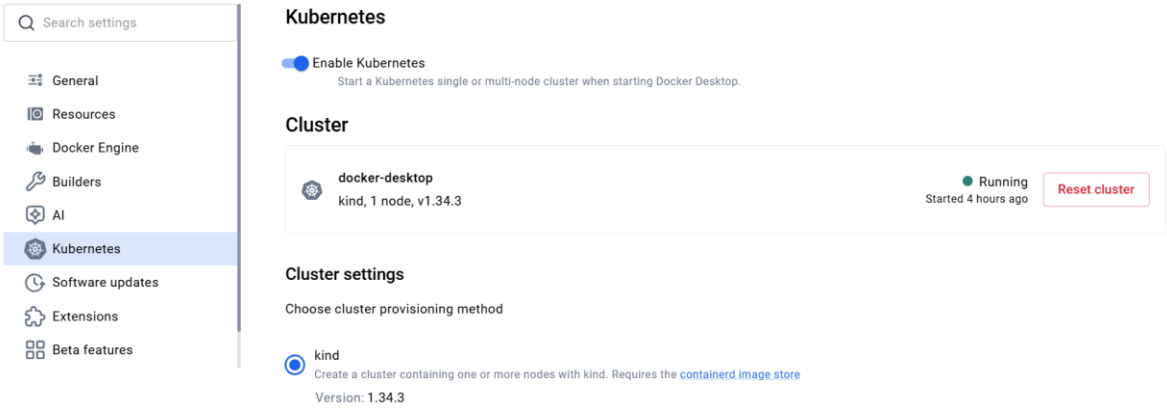
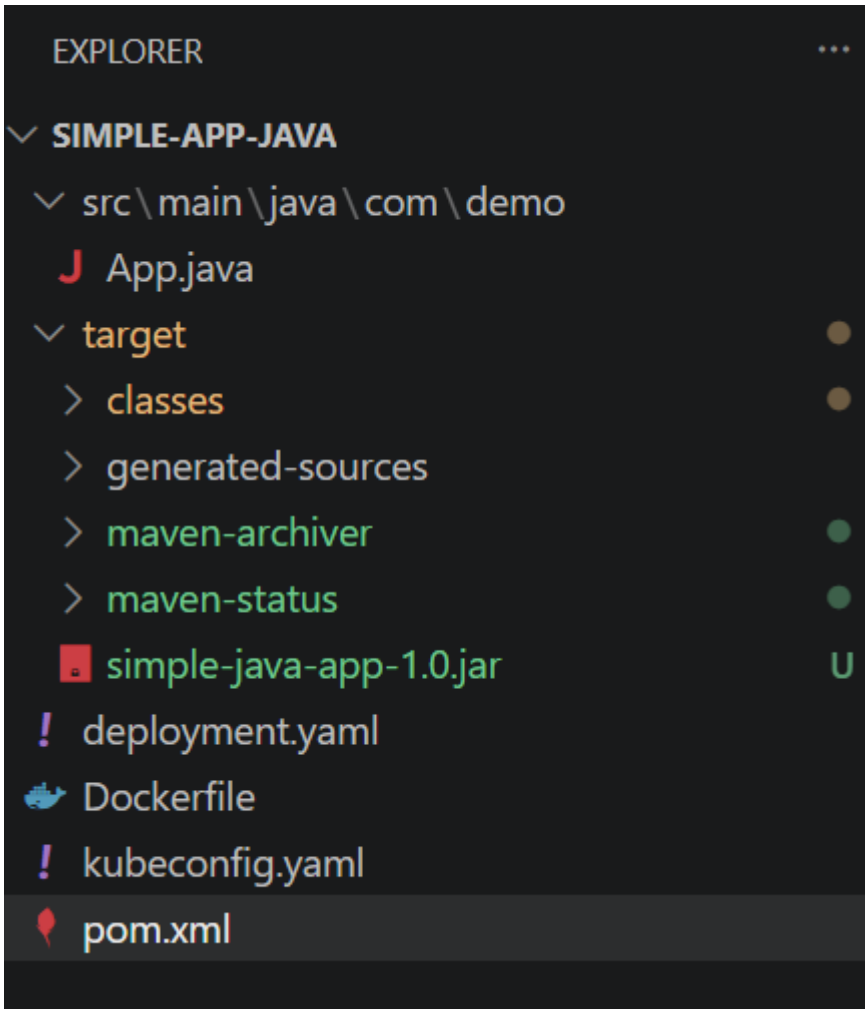
Jenkin – Docker – Kubernetes (CI-CD Demo)

Enable Kubernetes:



Create Simple Java Application:



App.java:

```java
package com.demo;

public class App {
    public static void main(String[] args) {
        System.out.println("Hello from Docker + Kubernetes!");
        while(true) {}
    }
}
```

Pom.xml:

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.demo</groupId>
    <artifactId>simple-java-app</artifactId>
    <version>1.0</version>

    <build>
        <plugins>
            <!-- Compiler -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.10.1</version>
                <configuration>
                    <source>17</source>
                    <target>17</target>
                </configuration>
            </plugin>

            <!-- Make executable JAR -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.2.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.demo.App</mainClass>
                        </manifest>
                    </archive>
```

```
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

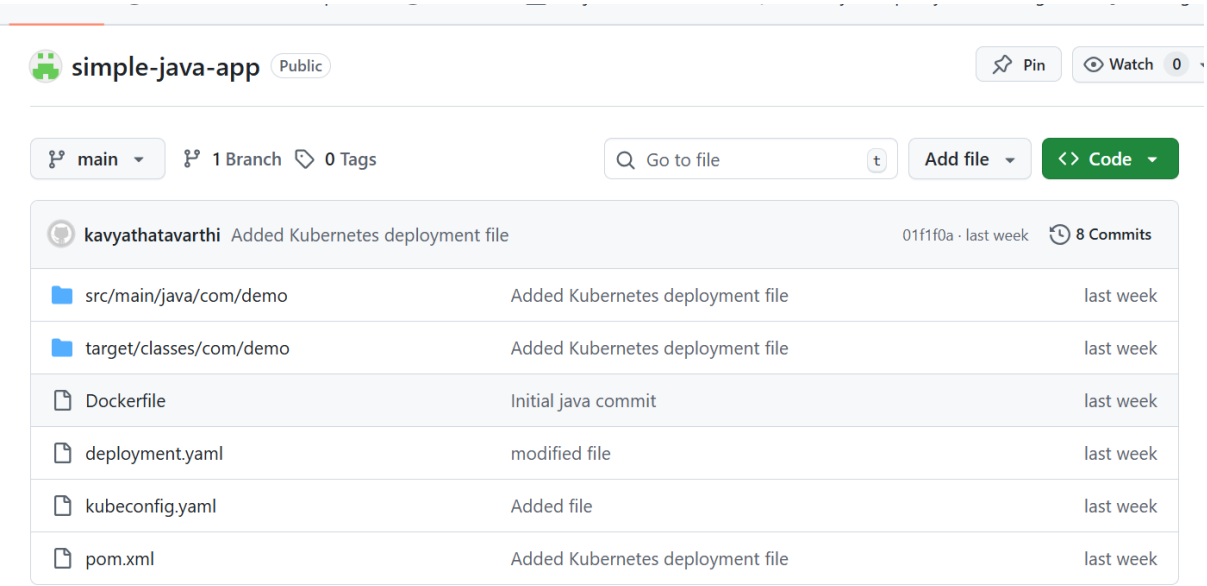
Deployment.yaml:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-app
spec:
  replicas: 1
    matchLabels:
      app: java-app
  template:
    metadata:
      labels:java-app
    spec:
      containers:
      - name: java-app
        image: shrivasha/java-app:latest
        - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: java-service
spec:
  type: NodePort
  selector:
    app: java-app
  ports:
  - port: 80
    targetPort: 8080
    nodePort: 30008
```

DockerFile:

```
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/simple-java-app-1.0.jar app.jar
CMD ["java", "-jar", "app.jar"]
```

Push to GitHub:



Configure Jenkins:

```
pipeline {
    agent any

    enviromment {
        DOCKER_IMAGE = "shrivarsha/java-app:latest"
    }

    stages {

        // ☑ FIXED: Kubernetes check with kubeconfig
        stage('Check Kubernetes') {
            steps {
                withCredentials([file(credentialsId: 'kubeconfig', variable:
'KUBECONFIG')]) {
                    bat 'kubectl cluster-info'
                    bat 'kubectl get nodes'
                }
            }
        }

        stage('Clone Repository') {
            steps {
                git branch: 'main',
                url: 'https://github.com/shrivarsha/simple-java-app.git'
            }
        }
```

```
        stage('Build with Maven') {
            steps {
                bat 'mvn clean package'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat "docker build -t %DOCKER_IMAGE% ."
            }
        }

        stage('Push to DockerHub') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'DOCKER_USER',
                    passwordVariable: 'DOCKER_PASS'
                )]) {
                    bat """
                    docker login -u %DOCKER_USER% -p %DOCKER_PASS%
                    docker push %DOCKER_IMAGE%
                    """
                }
            }
        }

        // ☑ FIXED: Kubernetes deploy with kubeconfig
        stage('Deploy to Kubernetes') {
            steps {
                withCredentials([file(credentialsId: 'kubeconfig', variable:
'KUBECONFIG')]) {
                    bat 'kubectl apply -f deployment.yaml'
                }
            }
        }
    }

    post {
        success {
            echo 'CI/CD Pipeline executed successfully!'
        }
        failure {
            echo 'Pipeline failed. Please check logs.'
        }
    }
}
```

Jenkins / Java-CICD / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline script

Script

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "shrivarsha/java-app:latest"
    }

    stages {

        // ✅ FIXED: Kubernetes check with kubeconfig
        stage('Check Kubernetes') {
            steps {
                withCredentials([file(credentialsId: 'kubeconfig', variabl
                    bat 'kubectl cluster-info'
                    bat 'kubectl get nodes'
                }
            }
        }
    }
```

☑ Use Groovy Sandbox

Jenkins / Java-CICD

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Stages
- Rename

Java-CICD

Permalinks

- Last build (#7), 7 days 4 hr ago
- Last stable build (#7), 7 days 4 hr ago
- Last successful build (#7), 7 days 4 hr ago
- Last failed build (#6), 7 days 4 hr ago
- Last unsuccessful build (#6), 7 days 4 hr ago
- Last completed build (#7), 7 days 4 hr ago

Jenkins / Java-CICD / #7 / Stages

#7

Rerun

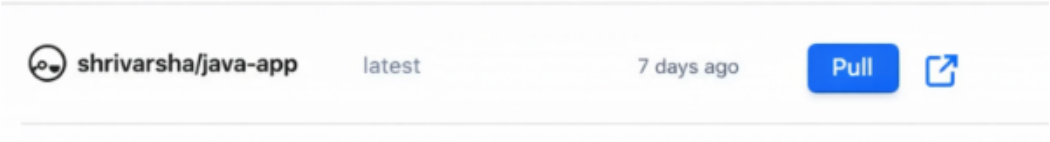
Started by kavyathatavarthi | Started 7 days 4 hr ago | Queued 6 ms | Took 32 sec

Graph

Start → Check Kubernetes → Clone Repository → Build with Maven → Build Docker Image → Push to DockerHub → Deploy to Kubernetes → Post Actions → End

```
C:\ProgramData\Jenkins\.jenkins\workspace\Java-CICD>kubectl apply -f deployment.yaml
service/java-service created
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Verify Docker and Kubernetes Deployment:



Kubernetes:

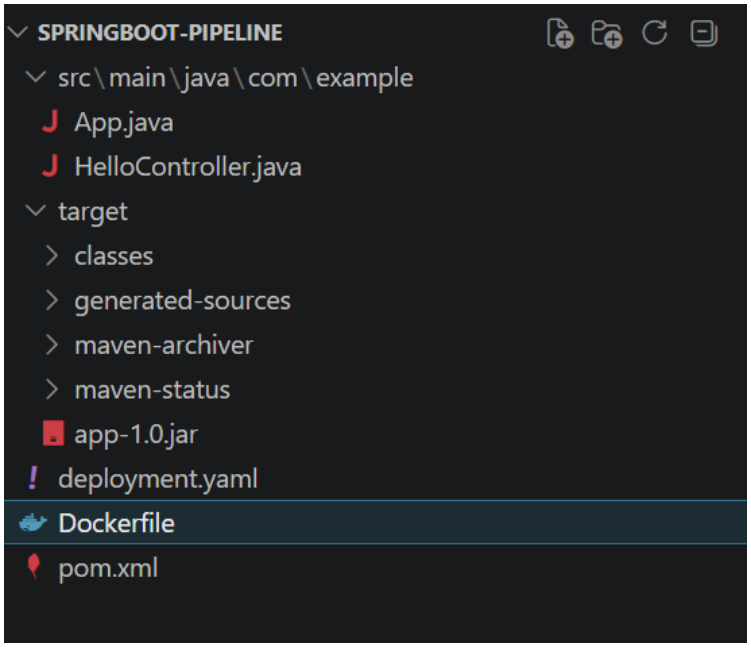
```
C:\agile\springboot-pipeline>kubectl get svc
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes       ClusterIP   10.96.0.1       <none>        443/TCP          5d
```

Spring Boot CI/CD Pipeline (GitHub + Jenkins + Docker + Kubernetes)

Install Tools:

```
C:\Users\shrivarsha> java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)

C:\Users\shrivarsha> mvn -version
Apache Maven 3.9.12 (848fbb4bf2d427b7bd2471c2fcd7ebd9a7a1)
Maven home: C:\Program Files\apache-maven-3.9.12
Java version: 25.0.2, vendor: Oracle Corporation, runtime: C:\Program Files\Java\25.0.2
Default Locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

C:\Users\shrivarsha> docker --version
Docker version 29.3.1, build c2be9cc

C:\Users\shrivarsha> docker ---version
Docker version 29.3.1, build c2be9cc

C:\Users\shrivarsha>kubectl version -client
Client Version: v1.34.1
Kustomize Version: v5.7.1
```

File Structure:



App.java

```java
package com.example;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class App {
    public static void main(String[] args) {
        SpringApplication.run(App.class, args);
    }
}
```

HelloContoller.java:

```java
package com.example;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    @GetMapping("/")
    public String home() {
        return "App Running";
    }
}
```

Pom.xml:

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>app</artifactId>
    <version>1.0</version>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.2.5</version>
    </parent>

    <properties>
```

```
        <java.version>17</java.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
    </build>

</project>
```

Deployment.java:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: springboot-app

spec:
  replicas: 1

  selector:
    matchLabels:
      app: springboot-app   # MUST MATCH BELOW

  template:
    metadata:
      labels:
        app: springboot-app   # MUST MATCH ABOVE

    spec:
      containers:
      - name: app
        image: springboot-app
        imagePullPolicy: Never
        ports:
        - containerPort: 8080
```

Dockerfile:

```
FROM eclipse-temuri
COPY target/*.jar app.jar
ENTRYPOINT ["java","-jar","/app.jar"]
```

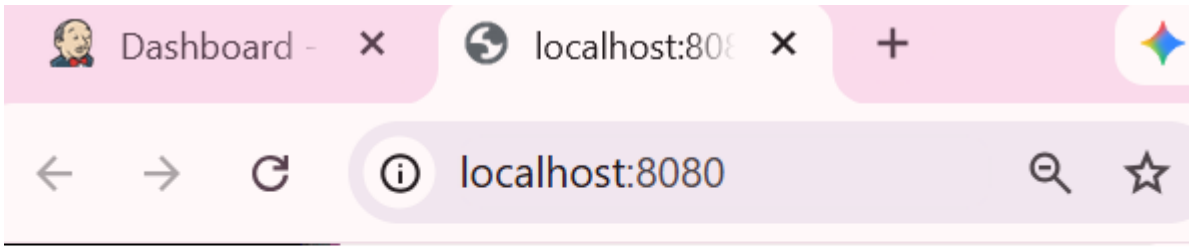
TargetFile:

```
C:\agile\springboot-pipeline>mvn clean package
[INFO] Scanning for projects...
[INFO]
```

```
[INFO] --- resources:3.3.1:testResources (default-testResources) @ app ---
[INFO] skip non existing resourceDirectory C:\agile\springboot-pipeline\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ app ---
[INFO] Building jar: C:\agile\springboot-pipeline\target\app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.369 s
[INFO] Finished at: 2026-04-10T19:30:00+05:30
[INFO] ------------------------------------------------------------------------
```

Run Locally:

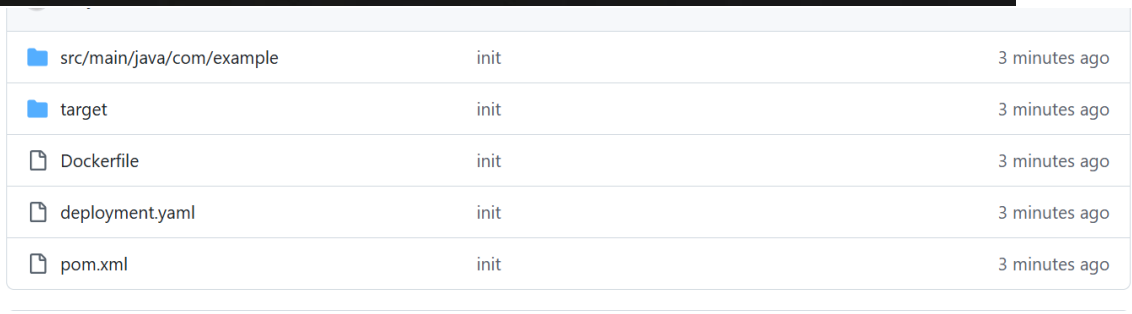
```
C:\agile\springboot-pipeline>mvn spring-boot:run
```



App Running

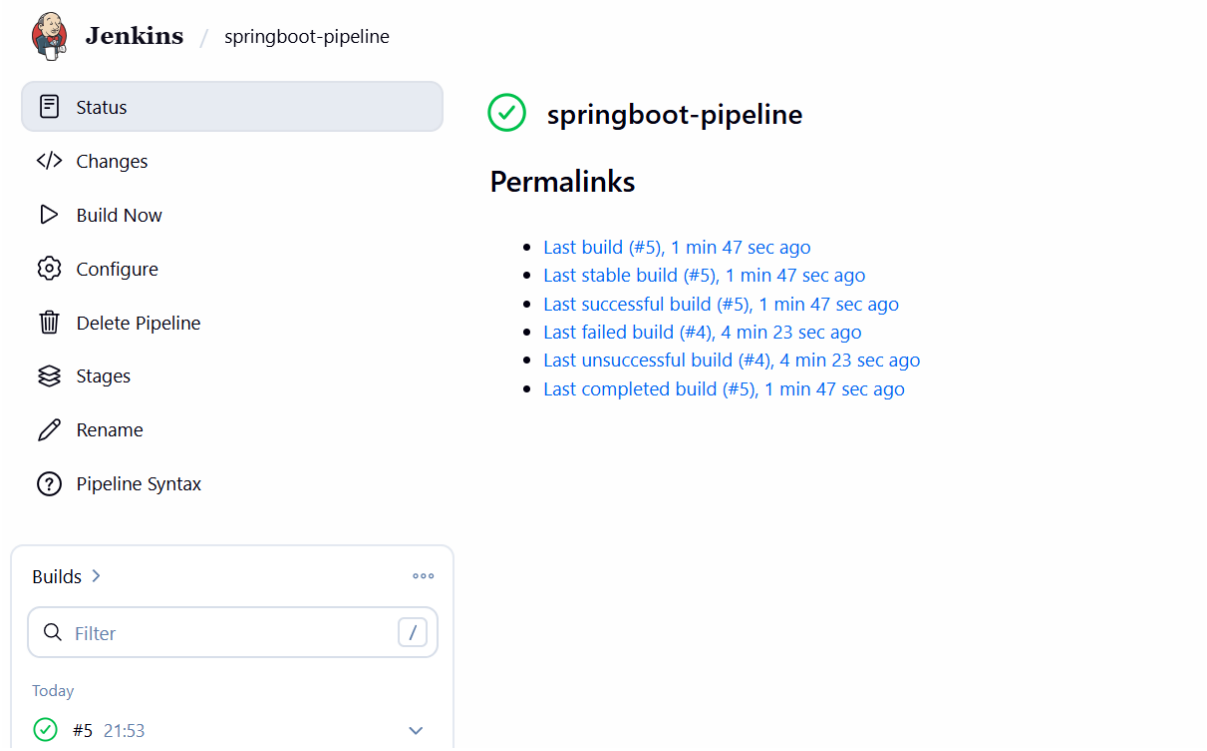
GitHub:

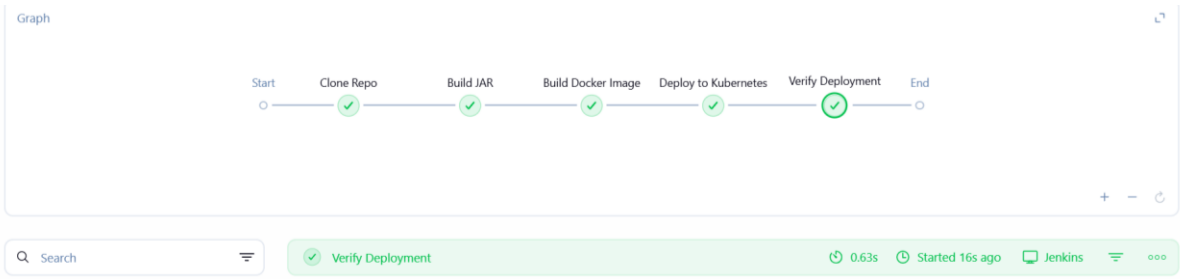
```
PS C:\agile\springboot-pipeline> git branch -M main
PS C:\agile\springboot-pipeline> git push -u origin main
Enumerating objects: 28, done.
Counting objects: 100% (28/28), done.
Delta compression using up to 12 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (28/28), 16.99 MiB | 949.00 KiB/s, done.
Total 28 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shrivarsha/springnboot-pipeline.git
* [new branch]  main -> main
branch 'main' set up to track 'origin/main'.
```



src/main/java/com/example	init	3 minutes ago
target	init	3 minutes ago
Dockerfile	init	3 minutes ago
deployment.yaml	init	3 minutes ago
pom.xml	init	3 minutes ago

Jenkins Pipeline:






```
C:\agile\springboot-pipeline>kubectl get pods
NAME                          READY   STATUS    RESTARTS      AGE
lms-v1-7788f54887-zq947       1/1     Running   2 (14m ago)   4d21h
lms-v2-549bbd7468-vk8xz       1/1     Running   2 (14m ago)   4d21h
lms-v3-7b69ffd6cc-cqzkh       1/1     Running   2 (14m ago)   4d21h
springboot-app-9b49d4ff8-7kcvr   1/1     Running   0             38s
```

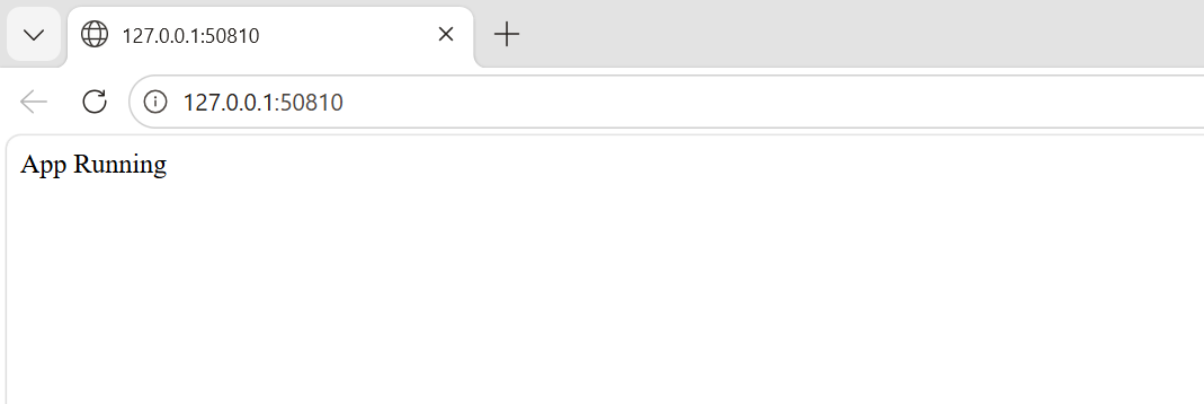
```
C:\agile\springboot-pipeline>minikube service springboot-app
```

NAMESPACE	NAME	TARGET PORT	URL
default	springboot-app	8080	http://192.168.49.2:31077

```
* Starting tunnel for service springboot-app.
```

NAMESPACE	NAME	TARGET PORT	URL
default	springboot-app		http://127.0.0.1:50810

```
* Opening service default/springboot-app in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



THANK YOU